Behavioral Observation and Hypothesis Testing

BIOL 1750 Laboratory Exercise

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Learning Objectives

By the end of this exercise you should be able to:

- Develop a testable behavioral hypothesis.
- Use a standardized ethogram to classify animal behavior.
- Evaluate inter-observer reliability.
- Compare scan sampling and all-occurrence sampling.
- Use a chi-square test to evaluate behavioral data.
- Interpret how sampling method influences scientific conclusions.

Video Feed Choices

San Diego Zoo https://zoo.sandiegozoo.org/live-cameras

Lincoln Park Zoo https://www.lpzoo.org/live-cams/

Smithsonian National Zoo https://nationalzoo.si.edu/webcams

Explore.org Live Feeds https://explore.org/livecams/currently-live

Background

Behavior is one of the most visible ways that animals interact with their environment. Because behavior changes continuously, biologists use standardized observation methods to collect objective and repeatable data.

An **ethogram** is a list of clearly defined behaviors used to classify observations. Before collecting behavioral data, researchers often compare observations between multiple observers to ensure that everyone is recording behaviors consistently.

In this exercise you will:

1. Practice using a standardized ethogram.
2. Measure inter-observer reliability.
3. Test a behavioral hypothesis using scan sampling.
4. Compare scan sampling with all-occurrence sampling.
5. Interpret how different sampling methods influence scientific conclusions.

Part I – Standardized Ethogram

Use the following ethogram throughout this exercise.

Code	Behavior	Definition
RST	Resting	Sitting, lying, or standing with little movement
MVE	Locomotion	Walking, running, climbing, swimming, or flying
FED	Feeding	Searching for, manipulating, or consuming food
GRM	Grooming	Self-maintenance such as grooming, scratching, or preening
SOC	Social Interaction	Direct interaction with another individual
EXP	Exploration	Investigating the environment or objects
OTH	Other	Behavior not fitting another category
OTS	Out of Sight	Animal cannot be observed

Part II – Inter-Observer Reliability

- ☐ Working with your partner, watch the same animal for approximately **10 minutes**.
- ☐ Every **30 seconds**, independently record the behavior occurring at that instant.

> **!** Important
>
> Do **not** discuss observations while collecting data.

Observation Table

Scan	Your Record	Partner Record	Agreement?
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

Calculate Observer Agreement

N Scans	N Agreements	Percent Agreement
20		

Questions

1. Was your observer agreement above 85%?

2. Which behaviors produced the most disagreement?

3. How could the ethogram be improved?

Part III – Develop a Hypothesis

For this exercise, your hypothesis should compare **two behavioral categories**.

> **Note**
>
> For the purpose of simplicity, we are really treating specific predictions like hypotheses today. A better hypothesis for each would be a broader biological statement about the relationship between two or more variables that we could support or refute using a number of more specific tests/predictions.

1. Fill out the null hypothesis implied by each of those below and take a moment to consider which behaviors from the ethogram you would fold into each category.
2. Select one of these 3 hypotheses to test.

Active vs. Inactive

Ha: *The animal spends more time active than inactive or the animal spends more time inactive than active.* **H0:**

Feeding vs. Not Feeding

Ha: *The animal spends most of its time feeding or the animal spends most of its time not feeding.* **H0:**

Social vs. Solitary

Ha: *The animal spends most of its time socializing or it spends most of its time not socializing.* **H0:**

State your hypotheses

Null hypothesis (H):

Alternative hypothesis (H):

Part IV – Scan Sampling

Protocol

You will observe one animal on your live animal camera for **15 minutes** and record which behavior the individual is engaged in every **30 seconds**, yielding a total **sample size of 20 scans**.
Note: You only need one observer per group for this.

- ☐ Before you begin, fill out the observation table below with your expected frequencies.
  - *Given the hypothesis you chose, how many of the 20 scans do you expect to record each of the following behaviors?*
- ☐ Complete your 15-minute observation.

Scan	Behavior
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	

- ☐ For each behavior, calculate the deviation (d) between observed (o) and expected (e) scan frequencies and square the result.

$$d = (o - e)^2$$

Code	Behavior	Expected (e)	Observed (o)	Deviation ($d = o - e$)	d^2	d^2/e
1						
2						
Sum (^{2})	NA	-	-	-	-	

Result

03C700B2: ______

$df = 1$

p-value:______________________________

Decision:

☐ Reject H

☐ Fail to reject H

Part V – Questions

1. What was your dependent variable? (0.5 pts)

2. What was your independent variable? (0.5 pts)

3. What is your sample size? (0.25 pts)

4. Did your experiments have replication of treatments? (0.25 pts)